Parabolics, Borels and residue kernels in the $\mathrm{PSL}_3(4)$ catalogue

L100 literal structural certificate

This note identifies the actual subgroups printed in the 29 diagrams of Gottschalk and Leemans' fixed Halle preprint [1], pp. 7–13. The acquired complete coset catalogue and its source-node bijections are inputs. We do not repeat the subgroup or geometry census. The retained 21-point realization of $G = \mathrm{PSL}_3(4)$, its order 20160, all its literal permutations, and its verified coordinate identification with $\mathrm{PG}(2, 4)$ are explicit premises. The journal edition is not collated here.

For each catalogue geometry, H_i is the saved stabilizer of its base object of type i, $B = \bigcap_i H_i$, and K_i is the already certified kernel of H_i on that object's entire residue. These are literal subsets of the same ordered list of elements of G. There are 99 parabolic positions, 99 kernel positions and 29 Borels. Identical subsets, rather than equal orders or isomorphic groups, reduce these 227 positions to 77 atoms. Every source position retains its exact atom identifier. The certificate first checks that a stored generating list generates exactly its asserted subset.

Lemma 1. *The actual point or line stabilizers have structure $2^4 : A_5$. The actual unordered point-pair or line-pair stabilizers used here have structure $2^4 : S_3$, and an incident point-line stabilizer has structure $2^4 : A_4$. Each colon is paid by an actual complement.*

Proof. Choose coordinates with the fixed point represented by e_1. An element of its stabilizer has an $\mathrm{SL}_3(4)$ representative $\left(\begin{smallmatrix} a & u \\ 0 & C \end{smallmatrix}\right)$ with $a \det C = 1$. Multiplication by the scalar a^{-1} is permitted because $a^3 = 1$. It gives a unique representative with upper-left entry 1 and lower block in $\mathrm{SL}_2(4)$. Consequently this group is the split extension of the additive row space $\mathbb{F}_4^2$ by $\mathrm{SL}_2(4)$. The latter acts faithfully on its five projective points and is A_5. The dual calculation gives the line stabilizer. A flag imposes one fixed point in this five-object action, yielding A_4. A pair imposes an unordered two-set, whose stabilizer in A_5 is S_3.

These descriptions are joined to each actual H_i without assuming that coordinate and permutation labels agree. The check finds its invariant point, line, flag, or unordered pair in the retained projective model. For a fixed line ℓ, it defines N as the elements fixing all five points of ℓ, and L as the stabilizer within H_i of a specified point outside ℓ. Dually it uses a pencil and a line not through its centre. It verifies $|N| = 16$, a commuting involution basis of length four, $N \lhd H_i$, $N \cap L = 1$, and every element has a unique expression nl. The faithful five-object action of L is checked on every element; it is the entire A_5, its specified point stabilizer, or its specified two-set stabilizer. This proves the claimed splitting and the actual action, in addition to the abstract group name. □

Lemma 2. *The other large atoms are A_6, A_5 and $L_3(2)$, with the indicated faithful geometric actions.*

Proof. For each order-360 atom the checker finds a retained invariant hyperoval and checks all 360 induced permutations of its six points. They are distinct and even, hence are precisely A_6. For each order-60 atom it checks either an invariant oval or an actual five-line orbit. The 60 induced permutations are distinct and even, proving A_5 also for the dual-oval stabilizers. A point action is not silently substituted for a line action.

For each order-168 atom it finds a retained Baer subplane, checks the seven points and its seven triple lines, and verifies the unique-line axiom on every point pair. Every recorded action preserves all seven lines and is faithful. An automorphism of the Fano plane is determined by

an ordered noncollinear triple: the other four points are recovered by line intersections. Thus its order is at most $7 \cdot 6 \cdot 4 = 168$. The exhibited 168 permutations are its full automorphism group $\mathrm{GL}_3(2) = \mathrm{PSL}_3(2)$, as asserted. □

The small structures. All remaining atoms have order at most 72. Their certificates use the following complete normal forms; no small-group identifier or order-only recognition is used. Here D_8 and Q_8 both have order eight. An elementary abelian atom has a commuting prime-order basis and every coordinate word exactly once; a cyclic atom has an element of full order. For A_4 and S_4, conjugation on the four actual cyclic subgroups of order three is faithful and gives respectively all 12 even or all 24 permutations. The groups S_3 and $7:3$ have actual generators a, b, orders $(3,2)$ or $(7,3)$, and $a^b = a^2$ or a^4 as appropriate; all words $a^i b^j$ occur exactly once.

For D_8 use $|a| = 4$, $|b| = 2$, $a^b = a^{-1}$. For Q_8 use $|a| = |b| = 4$, $a^2 = b^2$ and $a^b = a^{-1}$. In both cases the eight words $a^i b^j$ are checked. The atoms $2 \times D_8$ and $2 \times Q_8$ additionally exhibit an involution z outside this eight-element subgroup which commutes with both generators; all sixteen words are distinct and exhaust the atom.

The three groups $3 : S_3$, $3^2 : 4$ and $3^2 : Q_8$ have an explicitly normal translation group $N = \langle a, b \rangle \cong \mathbb{F}_3^2$. The saved complement has order 2, 4 or 8, intersects N trivially, and its products with the nine coordinate words exhaust the group. Its conjugation matrices on (a, b) are checked literally. For order 2 the action is $-I$. For order 4 its generator squares to $-I$; its minimal polynomial $X^2 + 1$ is irreducible over $\mathbb{F}_3$, giving the specified cyclic-four action. For order 8 the matrices are exactly

$$\{I, -I\} \ \cup \ \{M \in \mathrm{SL}_2(3) : \mathrm{tr}(M) = 0\}, \tag{1}$$

the standard quaternion subgroup. This identifies the semidirect product action as well as its factors. The inversion extension of 3^2 can also be written $3 : S_3$: choose one cyclic translation factor as the normal subgroup and the other together with the inverter as the S_3 complement.

Theorem 3. *All 29 Borel names and all 99 parabolic names in the fixed preprint agree with the actual catalogue groups. All boxed kernel structures agree as well. The outer colon in diagram 2.3 requires the nonsplit qualification in Lemma 4; no group or geometry count changes.*

Proof. The preceding witnesses certify all 77 literal atoms. Independently transcribed names at each printed node are compared through the accepted source-node bijection, including nodes of equal size. The Borel names are listed below; the full 99-node table, with each actual parabolic, kernel, atom and extension certificate, is exported as `source_names.json`. It is a literal finite join, not a new classification by subgroup orders.

Diagram	B	Diagram	B	Diagram	B
2.1	$2^4 : A_4$	3.1	A_4	4.1,4.8	3
2.2,2.5	S_4	3.2	4	4.2–4.7	2
2.3	$3^2 : 4$	3.3,3.4	D_8	4.9,4.10	2
2.4	$7 : 3$	3.5	3	4.11	1
2.6,2.7	Q_8	3.6	S_3	5.1–5.4	1
		3.7	A_4		

□

Lemma 4. *Every recorded residue kernel is checked inside its actual parabolic. The split, direct and nonsplit extensions have the following explicit interpretations.*

Proof. The acquired kernel is rechecked to be contained and normal. For each nontrivial kernel the checker constructs its entire coset partition inside H_i and the literal quotient multiplication table. Identical pairs (H_i, K_i) share one certificate; 61 such pairs cover all 99 positions. A trivial kernel uses the identity map, without manufacturing a new quotient computation.

In diagram 2.1 the boxed 2^4 is exactly the translation radical of Lemma 1. In diagrams 2.2, 2.5, 3.1, 3.3, 3.4 and 3.7 the boxed 2^2 is a specific subspace of that radical. Its exact basis, a vector-space complement, and the actual Levi action are retained. Thus the unparenthesized printed expression is paid in the form $(2^2 \times 2^2) : L$, with the action on the *whole* radical. We do not assert a central direct factor of H_i or an invariant complement merely from this typography. The checker explicitly records whether its chosen vector complement is invariant.

In diagram 2.7 the actual Q_8 kernel has a commuting involution complement, proving the displayed direct product. In 3.2 the actual cyclic-four kernel has an elementary-four complement, proving $4 : 2^2$. In 3.5 the actual cyclic-three kernel has a disjoint S_3 complement, proving $3 : S_3$. These conclusions use the specified kernels, not just some isomorphic normal subgroups.

In the central nodes of 4.2–4.7 the boxed involution is central and every square belongs to it. The quotient is elementary abelian of order eight. The extension is nonsplit: a complement would commute with the central kernel and make the whole group elementary abelian, whereas its exhibited dihedral factor has elements of order four. This is consistent with the source's dotted 2.2^3 notation.

Finally diagram 2.3 has $H = 3^2 : Q_8$ and the actual kernel $K = 3^2 : C_4$ of index two. Every element outside K is checked to have order four; in particular none is an involution. Hence there is no complement of order two. This also follows symbolically by projecting to Q_8: an involution maps to 1 or its unique central involution, both in the cyclic-four image of K. The preprint's displayed outer colon in $(3^2 : 4) : 2$ must therefore be read with a nonsplit extension dot in place of that split colon. The corrected assertion is $(3^2 : 4).2 = 3^2 : Q_8$ with the specified kernel. It is an auxiliary structural-notation correction, not a refutation of the principal geometry classification. □

Verification and scope. All products above are products of the actual retained permutations. A separate witness validator checks the saved maps, normal forms, geometric factors, quotient tables and actions without searching for new witnesses. Damaged generators, action tables, kernels, complements and source associations are rejected. The development record retains the initial syntax failure and the unsuccessful point-only search for the dual-oval atom; the final edition handles the five-line action. The complete ambient list, the coset models, the kernel meaning and the source-node transport remain acquired inputs, with their earlier proofs and distinct reviews. This note supplies their missing internal structural identifications, not another ambient census or a new measurement. Root's earlier suggestions concerning geometric stabilizers were advisory; this component's literal code and proof are authored by the publication-recovery agent and await distinct review.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL$(3, 4)$, Halle preprint 99–30, fixed 22-page source, pp. 4, 7–13.

[2] J. H. Conway et al., *Atlas of Finite Groups*, Oxford University Press, 1985. Notation antecedent cited by [1]; no Atlas subgroup classification is used to identify these atoms.